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Robust Residual Correction via Selective Deployment for Time Series Forecasting

(/pdf?id=C5U25I5APG)

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29 Apr 2026 (modified: 28 May 2026) NeurIPS 2026 Conference Submission
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
Revisions (/revisions?id=C5U25I5APG)
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TL;DR: We propose CRC, a plug-and-play framework that deploys post-hoc residual corrections only when validation evidence indicates net error reduction over a frozen baseline forecaster.

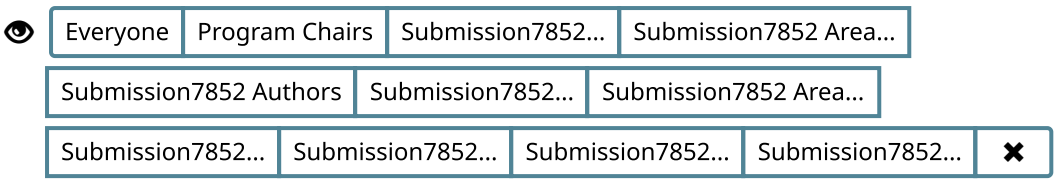
Abstract:
Post-hoc residual correction can reduce forecasting error, but always-on updates often fail to generalize: when residual signals are weak or noisy, expressive correctors can overfit and even increase the overall forecasting errors. We introduce CRC, a robust residual correction framework that couples a hybrid corrector with a validation-calibrated selective deployment rule. CRC proposes corrections via a conservative ridge "floor" plus conditional nonlinear refinement, but deploys them only when held-out evidence indicates net error reduction over a frozen baseline forecaster. This reframes correction as a decision problem: when to revise a forecast, not only how to fit residuals. Across standard long-sequence benchmarks and multiple backbones, CRC improves horizon-averaged MSE and MAE in most dataset-backbone pairs, with particularly large gains on complex datasets (e.g., over 20% relative MSE reduction on Traffic). We additionally report the non-degradation rate (NDR) as a stability diagnostic of harmful updates and ablate the validation-calibrated "firewall" that stabilizes MSE gains. Formal analysis motivating the design appears in Appendix A under idealized assumptions.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)
Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)
Contribution Type: General: Most submissions will fall into this type.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Understanding technical concepts
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2)
Submission Number: 7852

Discussion (?id=C5U25I5APG#discussion)

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Meta Review of Submission7852 by Area Chair asfz

Meta Review by Area Chair asfz 21 Jul 2026, 20:27 (modified: 24 Jul 2026, 03:30)

Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair asfz

Revisions (/revisions?id=dec225z0lL)

Metareview:

This submission proposes CRC, a plug-in residual corrector for frozen forecasting backbones that decomposes candidate corrections into a ridge "floor" plus a clipped, direction-gated MLP refinement, and its central idea deploys corrections selectively. A per-(variable,horizon) choice among ,baseline, +linear, and +linear+MLP is calibrated on validation and frozen at test time.

Strengths (consensus of reviewers):

- the "when to deploy, not only how to fit" framing is well motivated and practically relevant. The evaluation is broad. NDR is a useful diagnostic that averages obscure. Component ablations and sensitivity analyses (tau rescaling, validation budget) support the configuration. The paper is clear, honest about limitations, and code is released.

Weaknesses (summarized and augmented by the ACs question below):

- Uncertainty about novelty and baselines: See ACs follow-up question below.
- Missing statistics: No variability such as seed variance is reported despite averaging (raised by 14dv; vaUX/gnqC relatedly). Many Table 1 gains are quite small and the strongest-backbone (TimeXer) column includes two negative entries, so the "26 of 28 pairs" claim is currently weak without a more formal statistical analysis.
- Missing controls: The "always-on" control is missing (vaUX, gnqC): Table 2 removes filters one at a time but never shows the full corrector with no firewall, nor a simple validation-gated ridge, so the gain attributable to selective deployment itself is not isolated.
- Dependence on validation-test representativeness (MkCF, gnqC) deserves a discussion.

In summary, the paper has merit and addresses an important problem with the reviews being borderline accept and positive; however, there are currently non-trivial question/concerns that needs addressing before it can be considered for acceptance. The authors are encouraged to address the issues summarized above along with and the individual questions posed by the reviewers and follow up question included by the AC (below). In particular, if any of the reviewers' or AC's reservations are factually wrong, the authors should say so in the rebuttal.

AC follow-up questions to ensure an informative rebutal:

- AC-Q1: This is a very crowded area, and the novelty clarification needs a bit of clarification as far as I can tell. The paper seems to miss a few papers in the related work section, and possibly even as baselines, that appear highly relevant including TAFAS (AAAI 2025) which applies gated residual correction to frozen forecasters' per variable. PETSA and FAC use learnable per variable gates. SOLID adapts only when a residual-based detector warrants it. PIR's (cited) identification stage already modulates revision strength per instance as far as I can decode from the paper, so Section 4.3's descriptin of the different to PIR is not a big as descriped. Per-unit holdout selection among candidate forecasts is also classical (FFORMS/FFORMA).

TAFAS: Battling the Non-stationarity in Time Series Forecasting via Test-time Adaptation"

SOLID: "Calibration of Time-Series Forecasting: Detecting and Adapting Context-Driven Distribution Shift", https://arxiv.org/abs/2310.14838 (https://arxiv.org/abs/2310.14838)

PETSA: "Accurate Parameter-Efficient Test-Time Adaptation for Time Series Forecasting", https://arxiv.org/abs/2506.23424 (https://arxiv.org/abs/2506.23424)

FAC: "Towards Principled Test-Time Adaptation for Time Series Forecasting", 2026.

Based on these papers it seems like every part of CRC exists separately in prior work (gated correction of frozen forecasters, per-unit holdout selection a la FFORMA, non-degradation as a negative flip metric). Could the authors please help the AC by commenting on these papers and concisely explain what the novelty of the specific synthesis is? Also, it would be usefull to learn why the above methods have not been incldued as baselines.

- AC-Q2: Appendix A seems to oversell in that Prop A.3 is true by construction as far as I can see, and Prop A.4's displayed bound does not follow as stated (remark A.5 admits this), i.e., the abstract's "formal analysis" phrasing seems like a bit of a stretch. Could the authors comment, please?

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Author AC Confidential Comment by Authors

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by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))

25 Jul 2026, 12:01 (modified: 27 Jul 2026, 21:53)

Program Chairs, Senior Area Chairs, Area Chairs, Authors

Revisions (/revisions?id=veeG9lOwkw)

Comment:

Dear Area Chair and reviewers,

Thank you for the meta-review. We address meta-level concerns and AC questions below; per-reviewer detail is in separate posts.

Missing controls: always-on correction and validation-gated ridge

Reviewers noted missing always-on correction and a simple validation-gated baseline. **Table R1** (Gated Baseline Ladder), one batch: ETTh1/DLinear pl=96 & 336; Weather/TimesNet pl=96.

Config	Role	B1 NDR	B1 MAE
G0	Never update (Table 1 baseline)	—	0.4108
G1	Always-deploy*	**8.2%**	0.4738
G2	Ridge + val gate†	**97.9%**	0.4097
G5	Full CRC	**80.3%**	**0.3914**

*G1 uses validation training but **disables per-(variable, horizon) blend search** (fixed full weights)—this isolates **always-on deployment** of the corrector path, not “no validation.”
†G2 uses the **same Stage-A representation stack as G5** (MTL/STCL/priors); only the **deployment head** differs (no MLP; binary validation gate on the ridge path, w2=0).

(i) G5 vs. G1: in our ladder, selective per-(var,h) calibration is the key factor explaining the observed gains in both NDR and MAE (B1–B3 NDR ~8–34% vs ~78–80%; B1 MAE 0.4738 vs 0.3914).

(ii) G2 vs. G5: high NDR (97.9% B1) but weaker MSE/MAE than G5 (B1 MSE 0.3941 vs 0.3780; B2 G2 MAE 0.4693 vs G0 0.4654, G5 0.4392)—see gnqC/vaUX for full Table R1.

Multi-seed preview (B1): Predefined seed **2021** (not post hoc); G5 **0.3914** / **80.3%** NDR; reruns **0.3991 ± 0.0011**, G1 NDR **7.2 ± 1.5%**. **G1 vs. G5 trends** hold across seeds.

AC-Q1 — Novelty vs. TAFAS, SOLID, PETSA, FAC, FFORMA, PIR

Revision: expand Related Work with the clarification below (Methods/Limitations as needed).

Prior lines mainly target adaptation, revision magnitude, or model selection. **The distinction is not whether a gate exists, but what is gated:** CRC controls whether a **generated residual correction** should be trusted and deployed under a **frozen backbone** (per-(var,h), validation-calibrated at test).

Line	Primary decision	CRC emphasis
TAFAS, PETSA, FAC	How/whether to adapt the forecasting model or prediction process at test time	Frozen backbone + validation-calibrated deployment of post-hoc corrections
SOLID	Whether detected context shift should trigger adaptation	Per-(var,h) deployment from validation residuals + hybrid corrector
PIR	Which forecast instances to revise (identification + post-hoc revision)	Whether a generated correction should be trusted and deployed at (var,h); NDR as local-harm diagnostic
FFORMA / FFORMS	Which forecaster to ensemble	Which correction pathway within one frozen forecaster

CRC is **not merely a residual corrector**; it adds an explicit deployment decision layer **orthogonal to correction generation**—when and how residual corrections should be trusted and deployed. Per-(var,h) calibration and NDR expose local harm that global MSE hides.

Empirical distinction (Table R1, B1–B3): G1 vs G5 (always-deploy vs selective NDR/MAE); **G2 vs G5** (high NDR, weaker B1 MSE)—deployment ladder on frozen backbones, not TTA/PETSA protocol benchmarks.

We do not run TTA/FFORMA in rebuttal; we will discuss in revision and can add one frozen-backbone comparison if the AC recommends a protocol.

PIR (revision): soften §4.3—PIR **identifies which forecasts to revise** and performs post-hoc revision; CRC targets a different step: **whether a generated residual correction should be trusted and deployed** at fine (var,h) granularity (blend, ridge floor, NDR).

AC-Q2 — Appendix A / "formal analysis"

We agree Prop A.3 is largely by construction and Prop A.4 does not imply the displayed bound as stated (Remark A.5). Revision: present Appendix A as design intuition under idealized assumptions, not a deployment guarantee; soften abstract wording accordingly.

Statistics ("26/28") & validation–test representativeness

We agree seed variance and statistical reporting should be strengthened. **Revision:** mean±std over ≥3 seeds on representative settings; **paired statistical analysis on key comparisons** (including Table 3 CRC vs PIR)—we will **not** treat the Table 1 **26/28** pair count alone as significance.

PIR preview (≠26/28): 64 ETT cells (14dv): **40/64** MAE, **42/64** MSE wins; Wilcoxon MAE $p \approx 0.07$ (n.s. at 0.05). **PIR and CRC are complementary**, not MAE-dominance. Full Table 3 analysis in revision.

Validation–test mismatch: deployment rules are calibrated on validation; under shift, CRC tends toward conservative gates (high NDR, reduced MAE gain). We will elevate this in Limitations with G2 conservative behavior and stage-2 recalibration (MkCF/gnqC).

We will incorporate these revisions into the updated paper.

Sincerely,
The authors

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by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))

26 Jul 2026, 15:28 (modified: 26 Jul 2026, 15:29)

Program Chairs, Senior Area Chairs, Area Chairs, Authors

Revisions (/revisions?id=58fEfFr9AL)

[Deleted]

Official Review of Submission7852 by Reviewer vaUX

Official Review by Reviewer vaUX 24 Jun 2026, 11:19 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer vaUX

Revisions (/revisions?id=8T1RvdEK8G)

Summary:
The paper explores a situation in time series forecasting when an update to a model is being deployed and comments on the fact that the default mode of "always-on" update (so always accept the update) may lead to several problems. They propose a solution which is based on the fact that there is a set aside held-out set which can be consulted as to see if the new update would cause decrease in performance. Depending on this they effectively reframe the problem as a decision problem of "when to deploy". The results show that their method brings some improvements. There are however some limitations in the paper which are described in more detail below.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths

Motivation for the paper is good and well written. This sounds like an interesting problem and by reframing it to a "when to deploy" task sounds novel albeit perhaps simple (but if its elegantly simple and it works in this new framing that's a strenght on its own as well)

Weaknesses

Background

Background could be significantly expanded and better explained. There is not enough information given about the tasks of the datasets and also about the backbones. It would help the paper to feel less abstract, if the data and tasks solved by these methods would be better explained and illustrated. Additionally, when the description of these datasets is missing, one wonders about what happens with the inevitable real world anomalies (OOD data, sensor noise, extreme events) and this method - do the used benchmarks capture these or not? This is also linked to the language used - e.g. How to we "learn a correction" (does that mean that model is finetuned on more data?)? How do we "measure the improvement" (does that mean that we measure performance metrics on set aside data?)?

Results / simple baseline

In Table 1 and throughout the text a potentially simple baseline is missing - "always accept update". It is described in the motivation of the paper (when defining the identified gap), but then the paper itself never really compares against that simple approach. Unless this is what is meant by the "baseline" in Table 1, in which case the other simple approach (of "never update") is missing.

Results / limited comparisons

The NDR result give more insight into the results (over just the MSE numbers and % improvements), but they are missing in some places (e.g. Table 1).

Missing details

Not enough details are given about the held out calibration set. How was it made? Is it representative? Is it difficult to make (thus adding cost to using CRC?)?

Quality: 3: good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:
Major suggestion:

- Add missing baselines ("always update", "never update" vs the proposed "update when it's smart" which is your CRC method), or explain why they wouldn't make sense (if that's the case). Otherwise the claims made throughout the whole paper are not really founded on real evidence.

Minor points:

- Improve the background description (as in expanded on in the weaknesses). Also, while you briefly mention what PIR is in the Section 2, consider adding more details about how this method works here.
- Suggestion to restructure the paper so that there is a section called Results (right now its Experiments and Discussion directly). Maybe split it after 4.2. "Main result" and the following into it's own Results section

Limitations:
yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
None

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Author AC Confidential Comment**

Rebuttal by Authors

Edit

Rebuttal
by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))
24 Jul 2026, 18:11 (modified: 26 Jul 2026, 16:15) Program Chairs, Authors
Revisions (/revisions?id=xUgslEBW50)

Rebuttal:
Dear Reviewer vaUX,

Thank you for the review. We agree that the **"when to deploy"** framing is central, and that **missing explicit update policies** weakened the empirical story. We address this with **new controlled comparisons** (Table R1).

Table R1 setup. B1–B3 (ETTh1/DLinear @96 & @336; Weather/TimesNet @96); **G1–G5 from one unified batch** (same protocol).

Major suggestion — Never update / always update / selective update

Policy	Config	B1 MAE	B1 NDR	B2 NDR	B3 NDR
Never update	G0 (Table 1 baseline)	0.4108	—	—	—
Always accept update	G1 (always-deploy variant)	**0.4738**	**8.2%**	**10.6%**	**33.9%**
Selective update (CRC)	G5 (Full CRC)	**0.3914**	**80.3%**	**79.7%**	**78.5%**

Conceptually, Table 1 **baseline / CRC** rows are **G0 (never update)** and **G5 (Full CRC)**; Table R1 **instantiates** that comparison on **B1–B3** (fixed dataset–backbone–**single pred_len** each—not the horizon-averaged cells in Table 1, so MAE/MSE **need not match** Table 1 numerically). The missing control was **G1**: same corrector stack but **no per-(variable, horizon) blend calibration** (`w1=w2=1` everywhere; τ / direction gate still on). G1 shows that **always deploying** the corrector **hurts both MAE and NDR**, supporting the paper's motivation.

We also report **G2** and **G4** in the full ladder. **G2** (*same Stage-A as G5; ridge-only validation gate, no MLP*) reaches **97.9% / 97.1% / 90.0%** NDR—high safety—but **cannot match G5 on global MSE** (**B1 MSE 0.3941 vs. G5 0.3780**; **B2** G2 MAE **0.4693** vs. baseline **0.4654**, G5 **0.4392**). **G4** (light encoder, full blend) has **lower B1 NDR (62.1%)** than **G5 (80.3%)** while **G2** stays highest on safety; on **B2–B3**, G4 NDR is closer to G5 than G2. **G5** is **best on B3 MAE/MSE** (full Table R1 in appendix).

Background / datasets / anomalies

We agree the paper should better ground tasks and backbones. **Revision:** expand dataset descriptions (ETT hourly/minute, Weather multivariate, etc.), clarify that benchmarks use **standard cleaned splits** (not heavy OOD stress tests), and discuss sensor anomalies / drift in Limitations.

NDR missing in Table 1

We agree. **Revision:** add NDR to the main table or appendix. In rebuttal we report NDR alongside MAE/MSE for B1–B3 in Table R1 (see above).

Held-out calibration set

CRC uses the **same chronological validation split as Time-Series-Library** long-term forecasting (no test leakage). Stage-2 calibrates blend weights and τ on validation predictions only. The validation fraction for corrector training follows our sensitivity analysis in the paper. The backbone remains **frozen**; cost is one stage-2 pass after backbone training.

Minor (PIR, structure, limitations)

We will expand PIR in Related Work, add a dedicated **Results** subsection after §4.2, and strengthen Limitations per your checklist.

Thank you again for the constructive feedback.

Official Review of Submission7852 by Reviewer 14dv

Official Review by Reviewer 14dv 19 Jun 2026, 23:17 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 14dv

Revisions (/revisions?id=NBJAk0L08s)

Summary:

This paper reframes post-hoc residual correction as a decision problem: not only how to fit residuals, but when a forecast should be revised at all. The standard approach stacks an auxiliary module on a frozen backbone to correct its residual error, applying the correction unconditionally at test time. The authors argue this always-on regime is fragile, since a correction that helps in aggregate can worsen the error in individual cases. They call this the corrector's dilemma: biases that are reducible on average must be shielded from noise amplification at deployment.

CRC takes the history and the baseline prediction as input and selects between a linear or both a non-linear and linear refinement path using held-out evidence. The data is segmented with a sliding window, then two branches produce a rich embedding Z (one extracting statistics like means, trends, and correlations, the other creating averages over pairwise embeddings based on correlation-based K-nearest neighbors over variables). A residual estimator decomposes the correction into a linear part (ridge) and a non-linear part (MLP), with the non-linear term added conditionally under a deployment policy that targets failure modes like sign flips and selection noise.

The policy accepts non-linear refinement only when its sign matches the linear correction, clips corrections with validation-calibrated thresholds, freezes each scalar correction to linear-only or hybrid based on validation, and deploys only when validation gains exceed a margin.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
The paper addresses an interesting and well-motivated problem, and treating correction as a conditional decision rather than an unconditional transformation is a useful reframing. The presentation is a clear strength, the structure is easy to follow, they provide a formal analysis, and the experiments are seeded. The architecture is properly ablated, and the NDR gains attributable to each component are convincing, making a strong case for the design choices. The evaluation is also broad, spanning multiple standard benchmarks and standard backbone architectures.

The main weaknesses are reporting-related rather than conceptual. Variability across seeds is not surfaced, and the statistical significance of the improvements over the baseline is not established (see Questions). Neither gap undermines the contribution, but addressing both would strengthen confidence in the reported gains.

Quality: 3: good
Clarity: 4: excellent
Significance: 3: good
Originality: 3: good
Questions:

- Can you add the variability across seeds?
- Are the differences between PIR and CRC statistically significant (Table 3)?

Limitations:
None

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
None

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Author AC Confidential Comment**

Rebuttal by Authors

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Rebuttal
by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))
24 Jul 2026, 18:10 (modified: 27 Jul 2026, 21:52) Program Chairs, Authors
Revisions (/revisions?id=L0eg9UKB7t)

Rebuttal:
Dear Reviewer 14dv,

Thank you for the strong support and clear summary of CRC. We address the two reporting gaps below.

Q1 — Variability across seeds

We agree. **Revision:** we will report **mean ± std** over ≥3 seeds for Full CRC and key baselines on representative dataset–backbone pairs (at minimum ETTh1/DLinear and Weather/TimesNet).

Rebuttal preview (B1; same protocol/hyperparameters as Table R1, separate multi-seed job): Table R1 is a **representative fixed-seed run** (G5/G1 MAE **0.3914** / **0.4738**; NDR **80.3%** / **8.2%**). The ladder seed is **predefined** (`fix_seed` 2021), not selected by performance. **Additional three-seed reruns** give G5 MAE **0.3991 ± 0.0011** and NDR **79.3 ± 6.3%**; G1 remains **7.2 ± 1.5%** NDR with much higher MAE—the **relative G1 vs. G5 trends** are consistent across seeds.

Config	MAE (mean ± std)	NDR (mean ± std)
Full CRC (G5)	**0.3991 ± 0.0011**	**79.3 ± 6.3%**
Always-deploy (G1)	0.4828 ± 0.0131	7.2 ± 1.5%

The **large G1 vs. G5 gap in MAE and NDR** is consistent across seeds—not a single-seed artifact. We will extend multi-seed reporting to additional settings in the camera-ready.

Q2 — Statistical significance vs. PIR (Table 3)

We will add **paired statistical analysis** across datasets and horizons (bootstrap 95% CI on MAE/MSE differences and/or Wilcoxon signed-rank tests), aligned with **Table 3's seven-dataset scope** in the revision.

Rebuttal preview (ETT subset, finer granularity than Table 3): Table 3 reports **horizon-averaged** Δ over **28 dataset–backbone pairs** (7 datasets × 4 backbones). As an interim paired check on **64 cells**—**ETTh1, ETTh2, ETTm1, ETTm2** × 4 backbones × horizons {96, 192, 336, 720} (same frozen-backbone protocol as our internal PIR/CRC logs)—from a **Jul 2026 paired rerun**, CRC wins **40/64** (MAE) and **42/64** (MSE) with mean paired advantage to CRC (Δ=PIR−CRC > 0). **Sign tests** favor CRC on this subset (MAE p=0.03; MSE p=0.008). **Wilcoxon** signed-rank tests agree in direction (MAE p≈0.07—not significant at 0.05; MSE p≈0.003)—not uniform wins on every cell (~24 MAE cells still favor PIR). This **does not** replace Table 3 (e.g., ETTh1 is PIR-favored there under horizon averaging; Weather/Electricity/Traffic are outside the 64-cell subset). We **do not** claim CRC dominates PIR on all settings; we view **PIR and CRC as complementary** (instance-level **revision identification** vs **correction trustworthiness / deploy-or-reject** at (var,h)) and emphasize **NDR / selective deployment** (not in PIR logs).

Revision: report **per-cell** significance, effect sizes, and uncertainty on the **full Table 3 grid** (7 datasets). **PIR identifies which forecasts to revise and performs post-hoc revision; CRC decides whether a generated correction should be trusted and deployed** at (var,h), with **NDR** and validation-calibrated selective deployment—dimensions our PIR runs do not report—so Table 3 alone does not capture the full comparison. Small effect sizes will be stated explicitly, consistent with your view that the remaining gaps are **reporting-related**, not conceptual.

Limitations

We appreciate your "None" on limitations; we will still expand validation–test mismatch and static graph priors in the revision (raised by other reviewers).

Thank you again.

Official Review of Submission7852 by Reviewer MkCF

Official Review by Reviewer MkCF 18 Jun 2026, 00:24 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer MkCF

Revisions (/revisions?id=mFrolHwJNr)

Summary:
This paper investigates the post-processing residual correction problem in time series forecasting. Traditional methods typically train a residual corrector after freezing the backbone forecasting model to correct prediction bias. However, existing methods often employ always-on correction, directly updating residuals regardless of whether correction is needed. This can easily lead to overfitting when residual signals are weak or noise is high, increasing test errors. To address this, this paper proposes CRC, transforming residual correction from a question

of "how to fit the residuals" to "when to deploy the correction." CRC first generates candidate residuals using a hybrid corrector, then uses a selective deployment rule calibrated on a validation set to determine whether to apply the correction, thereby reducing harmful updates and improving prediction stability.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:

Strengths:

1.The authors changed the focus of posterior residual correction from "how to fit the residuals" to "when to deploy the correction." This perspective is more reliable and better aligns with the stability requirements of practical applications than always-on correction.

2.The authors propose a CRC framework that combines direction-aware residual representation, ridge floor, conditional MLP refinement, and validation-calibrated deployment firewall to achieve more effective residual correction.

3.The authors propose using NDR as an auxiliary diagnostic indicator to compensate for the problem that MSE/MAE can only reflect average performance and is difficult to expose local degradation.

4.The paper validates its core argument through comparison with PIR and multi-component ablation: the performance gains of residual correction come not only from a stronger corrector, but also from the selective deployment rules of validation set calibration.

Weaknesses:

1.The CRC deployment rules are calibrated on the validation set and frozen into the testing phase. The problem is that this set of rules may fail if the validation and test sets are not distributed correctly.

2.CRC uses a KNN graph based on the correlation of training data to construct variable adjacency relationships, with a default K=5, and then uses this graph consistently during the validation and testing phases. This graph can only capture simple correlations and may not be able to represent nonlinear relationships, time-delay relationships, or dynamically changing relationships.

3.The experiments covered 7 long sequence prediction datasets and 4 backbones, but mainly standard benchmarks such as ETTh, ETTm, Weather, Electricity, and Traffic. I think the focus should be on testing issues such as online updates and strong distribution drift.

4.There are many hyperparameters and rules, and the migration cost is not low.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:
1.How often do deployment rules need to be recalibrated?

2.What are the criteria for choosing K=5 in KNN?

3.What is the relationship between NDR and MSE/MAE?

Limitations:
The author did not adequately discuss the limitations he used.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes
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Rebuttal by Authors

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Rebuttal

by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))

24 Jul 2026, 18:14 (modified: 26 Jul 2026, 15:37) Program Chairs, Authors

Revisions (/revisions?id=VAW0EYIdDc)

Rebuttal:

Dear Reviewer MkCF,

Thank you for the detailed review and for highlighting **when to deploy** as the core contribution. We respond to each point below.

Table R1 (single unified batch, same protocol): B1 ETTh1/DLinear pl=96; B2 pl=336; B3 Weather/TimesNet pl=96.

W1 — Validation–test mismatch

We agree deployment rules are calibrated on validation and **may fail under distribution shift**. We will **expand Limitations** prominently.

Our validation-ratio sensitivity analysis (paper) shows reasonable stability under varying val fractions. Under mismatch, CRC tends toward **conservative gates**—high NDR but reduced MAE gain. From Table R1, **G2** (*same Stage-A as G5; ridge-only validation gate*) on B2: **97.1% NDR** but MAE **0.4693** vs. baseline **0.4654**.

Recalibration (Q1): When the deployment context changes, recalibrate by **retraining stage-2 only** (backbone frozen)—e.g., a new validation slice or a new τ quantile. **Frequency:** when the **operational validation window** changes (new season or month of held-out data), not on every test batch.

W2 — KNN graph (K=5)

The adjacency uses **training-set correlation** with top-K neighbors (default **K=5**). It captures **static linear co-movement**, not dynamic lags—we will state this clearly as a **limitation**.

Sensitivity (B1 Full CRC, `--fix_seed` 2021; K=5 row aligned with Table R1 G5 B1):

K	MAE	MSE	NDR
3	0.3912	0.3768	78.8%
5 (default)	**0.3914**	**0.3780**	**80.3%**
7	0.3929	0.3796	74.2%

Across **K ∈ {3, 5, 7}**, MAE/MSE vary modestly (~0.002 MAE); **K=5** (paper default) matches our Table R1 G5 B1 and yields the **highest NDR** in this sweep. We will include this table in the revision appendix.

W3 — Online / strong drift benchmarks

We agree standard ETT/Weather/Electricity/Traffic splits do not stress **online drift**. We scope CRC as **post-hoc correction under frozen backbones** with a single calibration window; **online recalibration** is future work (Limitations).

W4 — Hyperparameters / migration cost

CRC is two-stage: (1) train or warm-start the backbone once; (2) train the corrector on validation predictions. Table R1 includes **deployment ablations** (G2: restricted gate on same Stage-A stack; G4 deploy-only encoder). We will add an appendix **migration checklist** (backbone checkpoint + corrector weights).

Q3 — NDR vs. MSE/MAE

NDR is the fraction of (horizon × variable) pairs where correction **does not worsen** local MAE vs. the backbone; **MAE/MSE** are global averages. **High NDR does not imply maximum MAE reduction.** Table R1: **G2** on B1 has NDR **97.9%** but MSE **0.3941** vs. **G5** NDR **80.3%**, MSE **0.3780**; **G1** has NDR **8.2%**, MAE **0.4738**.

Limitations (your comment)

We will substantially expand Limitations (validation–test mismatch, static graph, drift, hyperparameters).

Thank you again.

Official Review of Submission7852 by Reviewer gnqC

Official Review by Reviewer gnqC 08 Jun 2026, 16:15 (modified: 23 Jul 2026, 21:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer gnqC

Revisions (/revisions?id=o7bvUApILc)

Summary:
This paper introduces a post-hoc residual correction method for time series forecasting. Based on the observations that always-on correction methods can improve some predictions but hurt others, the paper's main idea idea is to avoid applying residual corrections everywhere. The proposed CRC combines a ridge-based correction, a lightweight nonlinear refinement, and a validation-calibrated deployment rule to decide where corrections should be used. The method is evaluated on several standard long-horizon forecasting datasets with multiple backbone models. The results show improvements in MSE/MAE in most settings, with additional analysis using non-degradation rate (NDR) to measure how often corrections avoid worsening local forecasts.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths

1.The paper addresses a practical and important problem. The post-hoc correction can be useful, but unsafe corrections may degrade already good forecasts. 2.The selective deployment presents a novel perspective. Treating correction as a "when to deploy" problem, rather than only a residual fitting problem, is an intuitive and useful framing. 3.The experiments cover several standard datasets and backbone models, and the comparison with PIR is a useful addition. 4.The ablation study helps show the effect of the ridge floor, nonlinear refinement, encoder, and deployment firewall. 5.The NDR is useful, since aggregate MSE/MAE can hide local degradations across variables and horizons. 6.The paper is generally clear and easy to follow. The codes are also available.

Weaknesses

1. CRC has several moving parts. The paper would benefit from a clearer distinction between the core selective deployment mechanism and stabilizing details such as clipping, direction gating, point-wise selection, and shrink-to-base blending.
2. Since validation-calibrated deployment is central to CRC, a comparison with simpler validation-gated baselines would strengthen the empirical story. Without this comparison, it is somewhat hard to tell how much of the gain comes from the full CRC design rather than validation-based selection alone.
3. CRC may depend on how representative the validation period is. A clearer discussion of validation-test mismatch and possible recalibration would make the scope more transparent.
4. NDR is useful, but its relationship with MSE/MAE could be explained more clearly, since high NDR does not necessarily imply the largest aggregate error reduction.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:

1. Since validation-calibrated deployment is a key part of CRC, I would like to better understand how much of the gain comes from the full CRC design rather than from validation-based selection alone.
2. I find NDR useful, but I was not fully sure how it should be interpreted together with MSE/MAE. For example, in what cases should a practitioner prefer a method with higher NDR but slightly weaker average error?

3. The validation-ratio sensitivity analysis is helpful. Could the authors also comment on what may happen when the validation period is not very representative of the test period? A short discussion of possible failure cases or recalibration would be useful.
4. CRC has a few components, and I think the paper would be easier to read if the authors more clearly separated the core idea from implementation details. In particular, which parts are essential to the selective deployment mechanism, and which parts are stabilizers that could potentially be replaced?

Overall, I view the paper positively. My main questions are about the source of improvement and the scope of the full CRC design.

Limitations:
Mostly yes. The paper discusses relevant limitations, including the fixed graph prior, conservative gating, and dependence on the validation period. I would suggest making the validation-test mismatch point slightly more prominent, since the deployment rule is central to CRC.

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
No problem

Code Of Conduct Acknowledgement: Yes
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by Authors (Flora D. Salim (/profile?id=~Flora_D._Salim1), Mingyue Cheng (/profile?id=~Mingyue_Cheng1), YUNCHENG HUA (/profile?id=~YUNCHENG_HUA2), Hao Xue (/profile?id=~Hao_Xue1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission7852/Authors))
24 Jul 2026, 18:15 (modified: 26 Jul 2026, 16:14) Program Chairs, Authors
Revisions (/revisions?id=nNwV1tBFnT)

Rebuttal:
Dear Reviewer gnqC,

Thank you for the thoughtful and constructive review. We appreciate your positive view of the **selective deployment** framing and the usefulness of **NDR**. Below we address each weakness and question with **new controlled comparisons** (Table R1 below) in addition to our ablation study.

Table R1 (Gated Baseline Ladder). Three benchmarks under a **single unified batch** (same protocol throughout): **B1** ETTh1 + DLinear, pred_len 96; **B2** ETTh1 + DLinear, pred_len 336; **B3** Weather + TimesNet, pred_len 96. **G0** = frozen backbone only; **G1** = always-on deployment*; **G2** = ridge + validation gate only†; **G4** = deploy-only (blend + τ, light encoder); **G5** = Full CRC.

* **G1** uses validation training but **disables per-(variable, horizon) blend search** (fixed full weights)—this isolates **always-on deployment** of the corrector path, not "no validation."
† **G2** uses the **same Stage-A stack as G5** (MTL/STCL/priors); only the **deployment head** differs (no MLP; binary validation gate on the ridge path, w2=0)—a **deployment ablation**, not a minimal linear-only pipeline.

W1 & Q4 — Core selective deployment vs. stabilizers

We agree that the paper should separate the **core deployment mechanism** from stabilizing details. We will add a subsection *"Core deployment vs. stabilizers vs. representation"* and revise Fig. 1 accordingly:

Layer	Components	Role
Core deployment	Ridge floor; τ from validation; **per-(node, horizon) blending** (including shrink-to-base weights (0,0))	Decides **where** to deploy correction
Stabilizers	Direction gate; hard τ clipping	Reduces unsafe nonlinear updates
Representation	MTL; STCL; multi-scale physics priors	Improves **how well** residuals are fit

- **G1 (always-deploy):** NDR **8.2% / 10.6% / 33.9%** (B1/B2/B3) vs. Full CRC **80.3% / 79.7% / 78.5%** — in these runs, selective deployment is the main safety mechanism.
- **G3 (no direction gate):** effect is **setting-dependent** (B1 NDR **81.8%** vs. G5 **80.3%**; B2 **79.7%** vs. **78.5%**); we classify the direction gate as a **secondary stabilizer**, not the primary deployment rule.
- **G4 (deploy-only):** B1 NDR **62.1%** vs. Full CRC **80.3%**; on B3 Full CRC achieves **best MAE/MSE** (**0.1903 / 0.1408** vs. G4 **0.2081 / 0.1584**) while G4 remains strong on NDR—representation mainly boosts **deployment quality** on B1, not uniformly on all metrics.

W2 & Q1 — Full CRC vs. validation-based selection alone

We agree this comparison is central. Summary of Table R1:

Method	Description	B1 NDR	B2 NDR	B3 NDR	B1 MAE
G0	Baseline only	—	—	—	0.4108
G1	Always-deploy (blend search off)*	**8.2%**	**10.6%**	**33.9%**	0.4738
G2	Ridge + val gate only†	**97.9%**	**97.1%**	**90.0%**	0.4097
G4	Deploy-only encoder	62.1%	77.8%	79.4%	0.4097
G5	Full CRC	**80.3%**	**79.7%**	**78.5%**	**0.3914**

1. **Per-(node,h) deployment calibration** is the key factor explaining the NDR/MAE gap between G1 and G5 in Table R1. G1 keeps validation training but **disables blend grid search** (fixed full weights); NDR collapses and MAE degrades (B1 MAE **0.4738** vs. G5 **0.3914**). This is stronger than Table 2's single-filter ablations—it isolates **always-deploy** behavior on the full corrector path.
2. **High safety alone is insufficient (G2 vs. G5).** Ridge-only validation gating yields **very high NDR** (90–98%) but remains **conservative** and **does not recover the accuracy gains of the full deployment strategy**: weaker B1 MSE than G5 (**0.3941** vs. **0.3780**); on B2, G2 MAE **0.4693** exceeds baseline **0.4654** (NDR **97.1%**) while G5 MAE **0.4392** improves over G0. Full CRC targets **balanced safety and accuracy**—**stronger MSE** with **moderate NDR (~78–80%)**.
3. **Full CRC combines deployment and representation.** G4 lowers B1 NDR (**62.1%** vs. **80.3%**); on B3, Full CRC yields the **strongest MAE/MSE** in the ladder.

We will include the full Table R1 in the revision (main text or appendix).

W3 — Scope of theoretical analysis

We agree. We will revise the theory section to state clearly that it provides **intuition under stated assumptions**, not a formal deployment guarantee. Main claims remain **empirical**, supported by Table R1 and NDR.

W4, Q3 & Limitations — Validation representativeness

We will **elevate validation–test mismatch** in Limitations.

CRC calibrates deployment on the validation window. If validation is not representative of test:

- **Expected behavior:** more conservative (0,0) blend weights → **NDR may stay high** (as in G2) but **average MAE gain shrinks**.
- **Mitigation:** validation-ratio sensitivity (Fig. in paper); **recalibration** by retraining stage-2 on a new validation slice (backbone frozen) or adjusting the τ quantile.

We will add a “failure modes” paragraph with G2 on B2: NDR **97.1%** but MAE **worse than baseline** (0.4693 vs. 0.4654).

W5 & Q2 — NDR vs. MAE/MSE

We will add a short *Practitioner’s guide*:

- **MAE/MSE:** global average accuracy.
- **NDR:** fraction of (horizon × variable) pairs where correction **does not worsen** local error vs. the backbone (**MAE at each unit**, ties count as non-degrading—same as §4.1).

High NDR does not imply maximum MAE reduction (Table R1):

- **G2:** B1 NDR **97.9%**, MAE **0.4097**, MSE **0.3941** (small MAE gain over baseline 0.4108).
- **G5:** B1 NDR **80.3%**, MAE **0.3914**, MSE **0.3780** (stronger MSE vs. G0/G2).
- **G1:** B1 NDR **8.2%**, MAE **0.4738**.

When is higher NDR preferred? Settings where **local non-degradation** matters (heterogeneous multivariate series, auditable pipelines). CRC targets both when possible; G2 illustrates an explicit **safety–accuracy trade-off**.

Questions (summary)

Q	Answer
Q1	Always-deploy vs. selective: **G1→G5**. **G2** (same Stage-A as G5, ridge-only gate): high NDR, **conservative**—weaker B1 MSE vs G5. Full CRC adds nonlinear + full blend for MSE/NDR balance.
Q2	Stated W1, Q2/3 as a trade-off triangle.
Q3	Limitations + real-data + G2 conservative example in B2.
Q4	Core = blend + τ + ridge; stabilizers = dir gate; representation = MTL/STCL/priors (**G4**).

Limitations (your comment)

We will make Table R1 (and match **more prominent**, since the deployment rule is central to CRC.

Summary

We hope Table R1 and the three-layer decomposition address your main questions. **G2** (same representation as G5, restricted deployment head) achieves high NDR but is **too conservative** to match Full CRC on MAE/accuracy. **G1** shows that **per-unit deployment calibration**—not merely training a corrector on validation—is what drives NDR in our ladder.

Thank you again for the helpful feedback.

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